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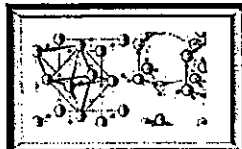
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## COLD FUSION TIMES

### QUALITY REPORTS CONTINUE TO CONFIRM COLD FUSION

**WELLESLEY, MA** - Several organizations, including US government agencies, have now reported confirmation of the cold fusion phenomena. In this issue of the **COLD FUSION TIMES**, we present reports which demonstrate the salient accumulation of new information from NASA, from Los Alamos, and from EPRI (the Electric Power Research Institute) consortium in California. This is remarkable both because it is corroboration by conservative, serious-minded scientists, which will spotlight cold fusion to receive further attention by the scientific community. Also this further increases the number of quality positive reports. Incredibly, even as this progress continues, the cold fusion skeptics and some hot fusion scientists, editors, and US Patent Office bureaucrats continue to deliberately stall development in this field; their hands still desperately cling to stale out-of-date newspapers and papers with scientific errors from 1989-90. Such stonewalling maliciously impedes progress in the field of energy production and nuclear physics within the United States. This may only be the second time in history that the United States has stopped scientific progress for political reasons. The first was the rebuff to Dr. Teller to develop the hydrogen bomb which would be reversed when the Soviets finally achieved detonation of an atomic bomb. >

### EPRI REPORTS POSITIVE DATA NUCLEAR PROCESSES IN DEUTERATED METALS

**PAL ALTO, CA** - The Electric Power Resource Institute's [EPRI] Report Summary on their "Development of Advanced Concepts for Nuclear Processes in Deuterated Metals" is available spanning 232 pages with detailed reviews of their meticulous work in the cold fusion field. The following is short excerpt of this important "quality-positive" technical report.

Since first announced in 1989 by Fleischmann, Pons, and Hawkins, "cold fusion" has been the subject of widespread interest and intense controversy. Palladium (Pd) cathodes electrochemically charged with deuterium (D) to unusually high D/Pd ratios exhibit episodes of heat in excess of measured electrical inputs. Although nuclear reaction products commensurate with the excess heat have not yet been observed, small but definite evidence of nuclear reactions have been detected at levels some 40 orders of magnitude greater than predicted by conventional nuclear theory.

The project team designed electrochemical cells that allowed precision calorimetry to be conducted while measuring all input heat from electrochemical and calibration resistor currents. The team operated 25 separate cell/calorimeters for periods of several days to several weeks each. Separately, the team operated 80 open cells to test various procedures for obtaining high cathode D/Pd ratios, a key condition for obtaining excess heat.

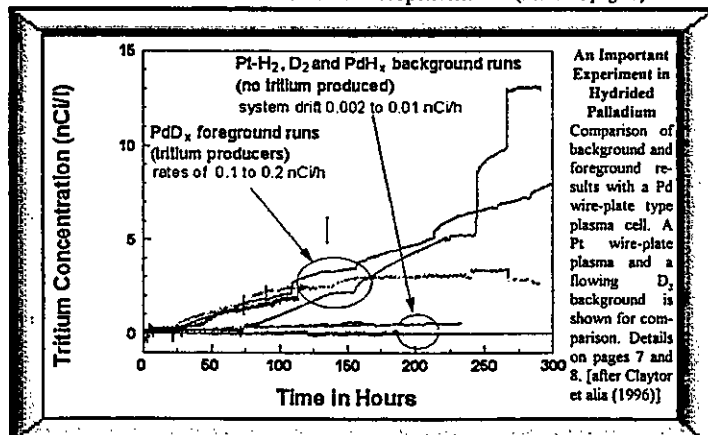
**Results** Three conditions were found characteristic of all cells yielding episodes of excess heat: (1) a D/Pd ratio less than 0.9, (2) initial appearance times of 8 to 23 days, and (3) cathodic current densities above 0.1 A/cm<sup>2</sup>. Excess powers ranging between a few percent to approximately 350% were observed, (continued on page 7)

### NASA CONFIRMS COLD FUSION IN LIGHT WATER/NICKEL SYSTEM

**CLEVELAND, OHIO** - NASA and a consulting company (NYMA, Inc.) have confirmed the cold fusion excess heat effect using light (ordinary) water with nickel and a Randall Mills-type setup. As reviewed in previous issues of the **COLD FUSION TIMES**, light water/nickel systems involve a variety of configurations, including those advocated by Dr. Mills using potassium carbonate electrolyte between an platinum anode and a nickel cathode. Drs. Niedra [NYMA] and Myers, Fralick and Baldwin [NASA] have prepared a technical memorandum (#107167) entitled "Replication of the Apparent excess Heat effect in a Light Water - Potassium Carbonate - Nickel electrolytic Cell". The report presents data which has confirmed excess enthalpies in the range of 6 to 68% for an electrolytic system fashioned after the Mills nickel prototypic cell. (More page 5)

### LOS ALAMOS NAT'L LAB REPORTS TRITIUM FROM CF

**NEW MEXICO** - As reported previously in the **COLD FUSION TIMES** (Vol 3, 2-3), Los Alamos National Laboratory has committed itself to continue research in this field. Los Alamos has now confirmed the generation of minute amounts of tritium in some cold fusion systems. Since there was negligible material contamination in their experiment, this is very important because there is only a nuclear source of this material. This distinguished paper has been long awaited by those in the field working in the enthalpic and helium production sectors of research and development. (More on page 7)



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# TOWARD THE FUTURE

Ed. - For years this column has focused on FUTURE events in cold fusion. Today, we look back through some harsh comments made by people who are so close to this field that their views deserve consideration. We respectfully disagree with some points made by Don Lancaster and Chris Tinsley, but these excerpts reflect the "pulse" of some in the field. A portion of the

criticism results from the difficulty in discerning useful systems from "New Age"-vapor systems.

Don Lancaster writes prolifically for *Electronics NOW* (<http://www.tinaja.com/libr01.html>). Here is a short excerpt from Vol. 66, 12-1995. Chris Tinsley writes for *Infinite Energy*. See references for more. ➤

## DOES PSEUDOSCIENCE STRIKES AGAIN?

Don Lancaster

Thatcher, Arizona -- "It's now our monsoon season out here in Arizona. For some weird reason, this seems to bring the perpetual-motion folks and pseudoscience enthusiasts out of the woodwork. I've recently been seeing one a day.

Meanwhile, all of the cold-fusion diehards appear to have gone into a "circle the wagons" state. They also seem to be running critically low

on ammunition. They are now centered on an *Infinite Energy* magazine and an CFNET online resource. The fact that they have now allied themselves with pyramid power (now renamed as tetrahedral superscalars) does not bode well.

Genuine new energy developments certainly will emerge. And research certainly should continue. So should independent thinking. For instance, the August 18th issue of *Science* tells us about a dramatic improvement in lower cost polymer solar cells on pages 920-921. These are still woefully inefficient and totally unstable, but they just got a whole lot better." ➤

## ON ITALIAN JUSTICE

Chris Tinsley

London, UK -- I've just heard the result of a court case in Italy, the one where Fleischmann and Pons and others sued the Repubblica newspaper for referring to cold fusion as scientific fraud. They lost the case. This has been gleefully reported on sci.physics.fusion by the scientific advisor to the defense, one Dr Douglas O Morrison of CERN - of whom my opinion is about as low as my opinion sinks. So, why am I pleased? Because I hope this will finally knock some sense into the heads of CF people. I said in Boston that we haven't got time for science in CF. That was not because I'm against science - I'm all for it - but because (as I also said) scientists will believe anything which they are funded to believe. Also, contrariwise. But there's only one way they'll be funded. If the CF wars are to be fought within science, then Planck's Other Constant will apply. Planck set this value at 20 years, the time it takes for fresh minds to overcome the past. ...

I will bet that some of the big companies who are now dabbling in the field are finding that the problems of reproducibility of results will not have gone away. I will also bet that some of them are cutting their losses and getting out of CF. The solutions to non-reproducibility are likely to be counter-intuitive.

A single corporation essentially means a single individual controlling its research. If that



individual has a fixed idea, then - if it is a wrong idea - no hope there. Meanwhile, out there we have all these little groups. ... Commercial secrecy is legitimate, but paranoia, greed and secretiveness are not.

I would suggest to these groups that they are far more likely to be relegated to the most microscopic of footnotes. Other people with a more open, co-operative and positive attitude will just walk past them. And let us be clear about the present role of science in CF. Its role is primarily to help in the development of high-performance machines, and to ensure the safety of products.

Once there are commercial products - even poor-quality, low performance products - then there will be no shortage of scientists to work on improving them. There will also be plenty of funding to find out the basic phenomenon. In the face of such funding, there will be plenty of believing scientists. To find the mechanism - or mechanisms - of CF is perhaps the most important aspect of the subject, but it is certainly the \*least\* urgent. ➤

[Ed. -- see page 4 for more on this matter of concern to some of the CF community.]



"One of the most interesting detailed and precise book available which extensively examines the very origins of gravitational science. Very highly recommended." **COLD FUSION TIMES**

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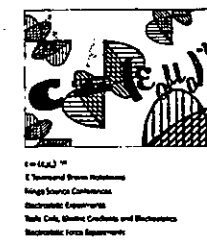
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## Photo Credits

Cover Page: after Clayton et al., description on page 7

Page 2: Chris Tinsley (*Infinite Energy Magazine*).

Page 3 & 4: Hal Fox, Douglas Morrison (Gayle Verner).

Page 5: Mills type set up (*Infinite Energy Magazine*).

Page 7: Dr. Clayton (*Fire from Ice*); VR site (JET Technology)

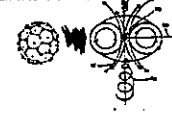
Page 9: Dr. Johnson (Winslow Martin *Tub*); Keanna Reeves (*20th Century Fox*)

Back Cover: [NASA] Egg Nebula [CRL2688, ~3,000 light-years away] in red light taken by the Hubble Space Telescope. A pair of "searchlight" beams emerges from a hidden red giant; they are crisscrossed by numerous bright arcs. The stellar ejection jets expand at 20 km/s (115,000 mph) and will only last 1,000 to 2,000 years (after R. Sahai and J. Trauger (JPL), WFP2 and NASA).

## American Cold Fusion Engineering

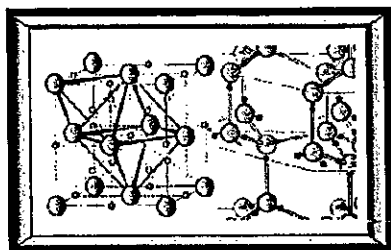
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<http://world.std.com/~mica/cft.html>

# PEOPLE IN THE NEWS

Ed. - The Third International Symposium on New Energy met in April 1996. Thirty-four papers were presented. Hal Fox, Editor of *New Energy News* and *Fusion Facts*, delivered the keynote address, and Eugene Mallove, of *Infinity Energy*, was the banquet speaker. James L. Griggs, Harold Aspiden and Dennis Cravens also presented. The following are selected excerpts; further details are in *New Energy News* (Vol. 4, No. 1, May 1996).

## REVIEW OF ISNE-3 AND NEW ENERGY TECHNOLOGIES

Hal Fox

Denver, Colorado --

Progress is being made in the development of three new energy technologies: cold nuclear fusion, high-density charge clusters, and super-magnets/super-motors -- providing thermal, electric, and mechanical energy. These three technologies are expected to form the basis for energy-producing systems of the 21st century, and all three are expected to be commercialized during 1996 and 1997.

At the ISNE-3 meeting, Dennis Cravens demonstrated a working power cell. Fox announced the successful invention, patent application, and early positive results of a new fiber power cell. This invention has the following inventors: Bruce Everingham, Avard Fairbanks, Hal Fox, Peter Gluck, and Shan Jin. Fox announced that specially plated nickel-fiber cathodes were the key that provides for ten to twenty times as much thermal output as compared with electrical input.

**Cold Fusion: New Energy Age** Eugene Mallove  
It is difficult to imagine a more profound reversal of scientific fortunes than what has been emerging



in the cold fusion field. One of the most reviled, disputed anomalies in the history of science is painfully, but inexorably heading toward acceptance by the scientific community. Not only that, but because cold fusion is rapidly entering the mainstream of media and industrial attention, it will have the collateral effect of focusing attention on other perhaps related anomalous excess energy machines -- New Energy devices that rely on vortex-cavitation principles and ones that are entirely electromagnetic, e.g. permanent magnet motors of unusual design that are claimed to be over-unity.

### The Patterson Power Cell Dennis Cravens

The Patterson Fuel Cell is a light-water, lithium electrolyte, nickel/palladium catalyst, which provides for a new type of water hydrolysis. The patented system has been demonstrated at prior conferences, was the main subject of a report on ABC's Nightline, and is considered by many experts as the leading contender for commercialization of New Energy devices.

### The Hydrosonic Pump James L. Griggs

The Hydrosonic Pump is a novel device for heating liquids by the use of shock waves, cavitation, or the implosion of bubbles within the liquid. The efficiency of the device has been shown to be significantly greater than 100%, and may be connected with the phenomenon of sonoluminescence, and the development of the Keeley motor by John Keeley in the late 1800s. The presentation included discussion on advantages, uses, and energy balances of the Hydrosonic pump, including the latest research findings.

### Vacuum Spin Energy Source Harold Aspiden

"Vacuum spin" consists of cylindrical or spherical pockets of aether spinning inside coextensive matter, and developing powerful electric fields. While

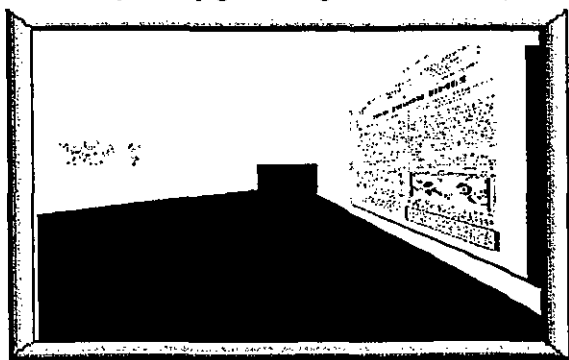
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## COLD FUSION -- now a VIRTUAL REALITY!!

Gayle Verner

Wellesley Hills, MA -- If you want to take a virtual walk on the wild side, see the **COLD FUSION TIMES GALLERIA** now open, and located on the World Wide Web. The **COLD FUSION TIMES** now has a Virtual Reality page which can be accessed through its URL at <http://world.std.com/~mica/cft.html>. For those who want to see the latest in virtual reality, or could never get enough time on the 3D headsets at electronics shows, here is a chance to view VR, and in 3D reread or return to the home page and just catch up on missed issues of THE most informative, up-to-date, solidly researched publication on the solid state fusion front.

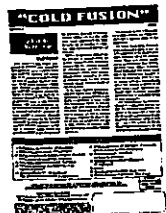
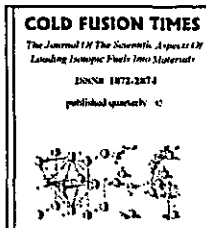
Top page editions of the **COLD FUSION TIMES** are posted on the walls of this virtual reality picture. As you hop through the colored corridors of this computer technology, watch the progress of CF in 3D, watch the Patterson Device, and other notables in the history of the field reported first in the **COLD FUSION TIMES**. Although this Virtual Cold Fusion World is not high resolution, it is a 3-Dimensional virtual world. The front pages, the Bose-condensates, ICCF issues, and even the development of the journal over several years, are resolvable. No one in this phenomenal field should miss the exciting development of virtual reality in web pages. Take a gander and see what you're missing.



Editor: ==> Software Requirement: VRML 1.0 browser such as Netscape 3.0 or other recent WWWbrowser. If you do not have the software, it is available by hypertextlink at the **COLD FUSION TIMES** home page under software section -- with loads of free telecommunication and networking software. If you want to see what VR is like, the figure is one view down a passageway.

OF GREAT INTEREST  
TO OUR READERS  
SEE FOR THE WORLD (PAGE 8)

3



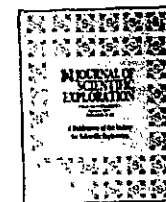
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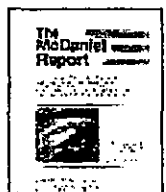
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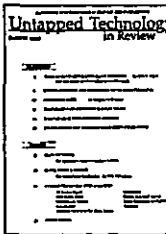


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# THE PATENT PAGE™ -- Analysis of Intellectual Property & Legal Issues

## MAD COW DISEASE? or JURY TAMPERING? -- YOU DECIDE

Italy - Many have read about the furor of "Mad Cow Disease" hitting the United Kingdom. But the way the Italian judiciary is making decisions about critical scientific matters, people may wonder if their court system wasn't infected too. A recently handed-down decision on whether Fleischmann and Pons were libeled was reversed. Outraged over this reversal of fact and the unfairness of the way the case was handled based upon more potential misstatements, many scientists are demanding justice, not to mention an investigation to vindicate the pair of scientists who have been corroborated by time and effort. Suspicion that something has gone terribly wrong focuses on a conspicuous bug in the ointment. Namely, CERN's physicist Dr. Douglas Morrison, longtime cold fusion opponent, whose stakes in keeping the phenomenon from gaining momentum, are legendary by now.

According to at least one report on the Internet, some of the opponents of cold fusion may have behaved like shameless self-promoters, in effect, "soliciting" his alleged well-honed-scientific skills to be used by the attorneys in the Italian courts. Somehow Dr. Morrison's self-appointed "Expert Witness from CERN" banner got the judges to swallow and endorse Morrison's erroneous heavily-biased statements. Should hot fusion physicists go out of their way to make themselves an "Expert Witness" and then influence the court? What is striking is the continued relatively successful effort of Dr. Douglas Morrison to attack cold fusion. This is more than an appearance of impropriety given the heavy conflict of interest.

However, even though it was the skeptics who sought out La Repubblica and then misstated facts, even though such agents of CERN and hot fusion continue to operate at a relatively low level of accuracy, in the interest of truth, the COLD FUSION TIMES now presents Dr. Morrison's own report with his CERN-anti cold fusion parallax view.

### CERN's REPORT ON COURT JUDGEMENT Douglas Morrison, CERN



CERN - Martin Fleischmann, Stanley Pons, Tullio Bressani, Guiliano Preparata, and Emilio Del Giudice Versus Editoriale La Repubblica s.p.a., Eugenio Scalfari and Giovanni Maria Pace. In October 1991, the La Repubblica published an article by their science editor, Giovanni Maria Pace, which reviewed the book "False Prophets" and essentially suggested that cold fusion

was "scientific fraud" and the names of Fleischmann and Pons were mentioned. Letters from Bressani (Turin), Preparata (Milan) and Del Giudice (Milan) were written to the newspaper and published. The five then sued the newspaper, its editor and the journalist for 8 billion lire (about \$5 million at that time). The court appointed Prof. Giovanni Licheri of Cagliari to assess the 105 pieces of scientific evidence presented. The five plaintiffs asked Prof. Daniele Gozzi (Rome) to be their adviser. La Repubblica asked Douglas Morrison of CERN, Geneva, to be their scientific adviser. See Nature 363(1993)107. Both sides submitted further written evidence. This may be the first time, or one of the first times that a law court has been asked to decide on a scientific issue.

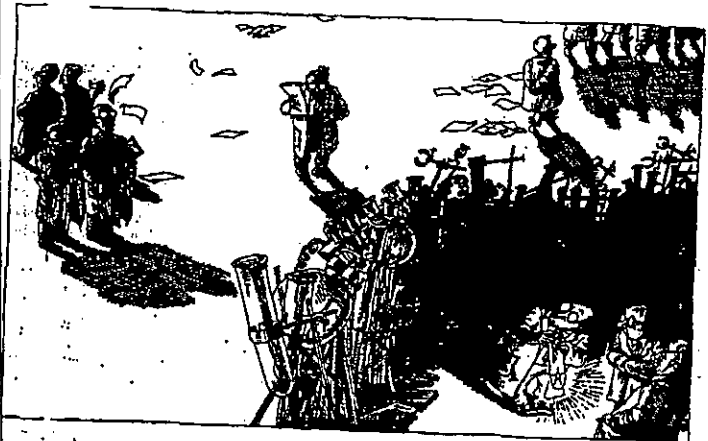
The decision of the court has just been received. It rejected the plaint of Fleischmann, Pons, Bressani, Preparata, and Del Giudice. It condemned the five to pay the legal expenses of 28 million lire (less than \$20 000) of the defendants.

Firstly the court considered whether the manner in which Fleischmann and Pons presented their results was correct. They concluded that it was correct because on the 13th



Spontaneous prion aggregation -- also known as Bovine spongiform encephalopathy (BSE) -- has been the near ruin of British cattle industry. BSE is blamed on prions which are aggregable proteins present in nervous tissue, and are shown above. It is unlikely that the Judges or skeptics who knock cold fusion have BSE, but given the plethora of positive reports now documented over seven years, then who knows?

## COLD FUSION TIMES' FAVORITE CARTOON OF THE "MONTH"



The secret life of cold fusion  
after the Economist September 30, 1989

March 1989 they submitted their paper for publication, and it was accepted on the 22nd March before their press conference of the 23rd March. However there were some negative aspects;

1. They failed to mention the work of Prof. Stephen Jones which began in 1986. Fleischmann and Pons had agreed with Jones to submit their papers for publication together on the 24th March
2. When it was pointed out to Fleischmann and Pons that their gamma ray peak was at an impossible energy, they changed the scale
3. They claimed to have observed and measured fusion for a long time but were not subject to gamma radiation - therefore it was not fusion
4. In April 1989, the US government set up a committee of 22 scientists to check the results. Despite complicated work, the results were negative. This is described in the book by John Huizenga, the Co-Chairman, entitled "Cold Fusion - the Scientific Fiasco of the Century".

5. On the 8th July 1989, in the Deseret News (daily newspaper published in Utah) appeared an article (with photograph) where Pons declared that he had made an apparatus of the size of a thermos which would satisfy the needs of a normal family and could make tea. Pons also said that the boiler was giving off 10 to 15 times the energy put in. The court noted that little progress had been made since 1991. There was no good theory to explain the claims of cold fusion and there was a failure to observe the products of fusion as would be expected (tritons, neutrons, protons, He, He and gamma rays) The court noted the comportment of Fleischmann and Pons who provided different and inconsistent data at different times, noted that they omitted to cite the work of Prof. Jones, noted the manner in which they dealt with the press, and noted how they considered future developments and concluded that they were separated from reality.

The court noted that most scientists had now abandoned cold fusion. Thus the court judged that Giovanni Maria Pace was justified in making his comments in the exercise of his profession, since there had been great discussion and contestation. The above is a brief summary of the 14 page judgement. Since this was translated from Italian, it is possible that there are some small errors in translation (please inform me so I may correct them) but the sense of the judgement is clear.

(Ed. - Should Court witnesses be pro-active in civil suits that effect their own livelihood, or would that make them more like "ambulance chasers"? Should so-called "Expert Witness" reports be accepted by the scientific community when the evidence from 1990-1996 is ignored? This report by Dr. Morrison of his own influence upon the Italian Court may have the appearance of bias but he remains a contributor to the involved case, and his parallax, albeit skeptical, point-of-view is relevant.)

### COLD FUSION TIMES

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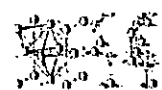
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# MATERIALS CORNER NASA REPLICATES NICKEL-BASED CF, OTHER NEWS FROM LOS ALAMOS, and from CHINA

## "Normal Temperature" FUSION in CHINA

Prof. Xing Zhong Li Dept. of Physics, Tsinghua Univ Beijing, China

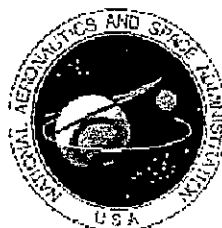
(continued from page 10) Since we had very little financial support, and we believed at that time that the nuclear signals should be commensurate with the "excess heat," then, the nuclear methods had higher research funding priority than the calorimetric methods. Gradually, we understood that "Thunder without Lightning" might be the feature of this "cold fusion" phenomenon, and the conventional nuclear signals might not be commensurate with the excess heat. So the calorimetric measurement might be the more suitable technique to identify this phenomenon. The preliminary results showed that there were some features which could be explained as the evidence of the "excess heat" (the turning point in the cooling curve, the sudden change of the temperature, etc.); however, more careful work is necessary before it becomes compelling evidence. The electrical discharge work has been improved since the Nagoya conference. Although the neutron energy was found to be 2.45 MeV (normal), the yield of neutrons was still found to be higher than normal (abnormal). In order to reduce the yield of the normal neutron, a careful study on electrical discharge was done to identify a set of operation parameters with which the abnormal neutron yield became more apparent. A new set of Si-Li detectors was used to refine the X-ray detection. A fine structure of X-ray spectrum was found definitely, and the difference between the discharge in deuterium gas and in hydrogen gas has been identified (the energy and the intensity of the X-ray for Nb-D system was higher than that for Nb-H system).

An interesting "X-ray radiation after death" was found after shutdown of the electrical glow discharge. It may last as long as 1,000 seconds. More careful work should be done before any possible explanation on this phenomenon is given. But the similarity to "heat after death" surprised all the attendees. A careful theoretical study was done to analyze the X-ray spectrum. It was found that among the nearly 200 spectra, about 100 spectra were abnormal. None of the applicable theories could explain the strange features in the X-ray spectrum. It was considered as evidence of the abnormality in the electrical discharge tube. A calorimetric measurement was done with the electrical discharge tube. A sudden change of the slope of the cooling curve (temperature versus time) was found in a series of discharges with different electrodes and gases. Ni-H system had been studied with the emphasis on "excess heat," however, a different conclusion was found. Although the Italian results had been reproduced, different explanations were given about this similar phenomenon. Based on the new explanation, the careful work proved that the "excess heat" in the Ni-H system disappeared.

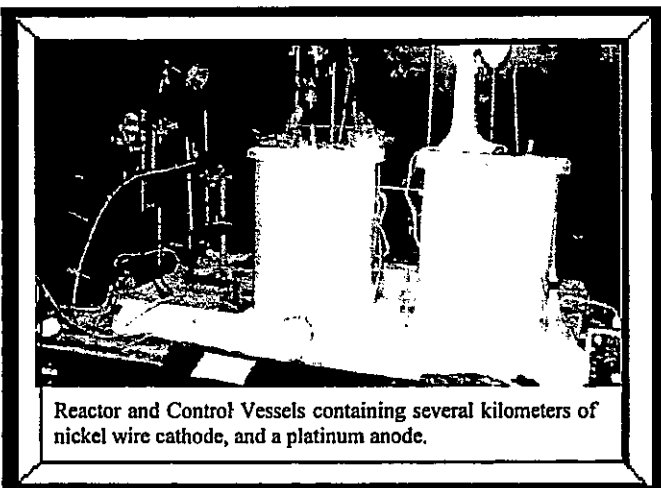
The gas-loading technique had been studied for more than five years in China in combination with the nuclear signal detection. At this symposium, a preliminary study on "excess heat" in a gas-loading system introduced interesting discussion. Since the "heat after death" phenomenon and other gas-loading experiments in Italy and in India revealed the possibility of observing the excess heat in a gas-loading facility, the advantages of being more cost effective, and safer in a gas-loading system became attractive. Besides, it had the feature of accumulating the reaction products in the reaction vessel which might be detected later, after a long period of operation. A pair of twin systems had been manufactured to see the difference between the deuterium-palladium system and hydrogen-palladium system. A sharp contrast had been observed already. More results were expected after further operations. ➤

## Replication Of The Apparent Excess Heat Effect In A Light Water - Potassium Carbonate - Nickel Electrolytic Cell

Janis M. Niedra [NYMA] and Ira T. Myers, Gustave C. Fralick and Richard S. Baldwin [NASA] Technical mem. (#107167)



Cleveland, Ohio - Replication of experiments claiming to demonstrate excess heat production in light water  $\text{Ni-K}_2\text{CO}_3$  electrolytic cells was found to produce an apparent excess heat of 11 W maximum, for 60 W electrical power into the cell. Power gains ranged from 1.06 to 1.68. The cell was operated at four different dc current levels plus one pulsed current run at 1 Hz, 10% duty cycle.



Reactor and Control Vessels containing several kilometers of nickel wire cathode, and a platinum anode.

The 28 liter cell used in these verification test was on loan from a private corporation whose own tests with similar cells are documented to produce 50W steady excess heat for a continuous period exceeding hundreds of days.

The apparent excess heat can not be readily explained either in terms of nonlinearity of the cell's thermal conductance at a low temperature differential or by thermoelectric heat pumping. However, the present data do admit efficient recombination of dissolved hydrogen-oxygen as an ordinary explanation. Calorimetry methods and heat balance calculations for the verification tests are described. Considering the large magnitude of benefit if this effect is found to be a genuine new energy source, a more thorough investigation of evolved heat in the nickel-hydrogen system in both electrolytic and gaseous loading cells remains warranted. ➤

## COMING ISSUES: COLD FUSION TIMES

What do you know about Lavoisier and the incredible rich history of South American platinum metallurgy? Astonishing technology. Stay tuned.



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## COLD FUSION TECHNOLOGIES

(continued from page 3) unseen ionization or polarization in matter, the spin does develop kinetic energies from an electrical coupling with outer space by the phase-lock of a quantum synchronization with the surrounding aether. The kinetic energy obtained with low electrical input power amounts to more than a billion joules per cubic meter -- thus providing an over-unity prospect. By regenerating the spin condition cyclically, one can access aether energy at a power rate determined by the pulse frequency and the physical volume of the 'vacuum spin' form. Independent experiments, and relationship to the so-called N-machine will be discussed.

### Brown's Gas A. Michrowski & Van Rhee

There is a demand for the design and realization of healthy, autonomous energy homes. The proposed modality for the safe supply of energy uses a solar energy system generating DC power for a stoichiometric 2:1 proportion hydrogen/oxygen gas (Brown's Gas) generating electrolytic cells. This gas generates and stores electrical power, feeds heating and cooling systems and provides clean water. This combination involves a relatively mature and safe technology and is available at present.

### The Beltrami Vector Field - Donald Reed

An exploration of the Beltrami vortex topology derived from the annals of hydrodynamics and electrodynamics of the 19th century - as a possible archetype for the primordial electromagnetic vacuum fields of space. The significance of this topology is under-scored in the contra-rotating non-linear vortex filaments discovered in plasma focus research. It is argued that chiral Beltrami morphology may be universal in microscopic quantum phenomena as well as macroscopic plasma levels. A new classical model for the self-action of the electromagnetic field, devoid of standard divergences attendant with QED analysis is surveyed, incorporating the Beltrami vortex morphology.

### Hadronic Energy Ruggero Maria Santilli

Outline of the new, subnuclear, clean Hadronic energy, and indications these new forms of energy generally contain the principles for recycling nuclear waste. The basis of the relativistic hadronic mechanics (RHM) is a quantitative representation of interactions due to mutual wave overlappings, which are nonlinear (in the wavefunction), nonlocal and nonpotential. They are therefore outside the domain of relativistic Quantum Mechanics. Importantly, nonlinear, nonlocal and nonpotential interactions can produce attraction among particles, i.e. nuclei or atoms of the same charge - a fact already established in the proton-proton bonding in the nuclear structure, superconductivity, and chemical bonding. RHM is expected to provide quantitative treatments of cold fusion and other new energy technologies, including RHM providing a rigorous explanation of how cold fusion processes can be initiated even against repulsive Coulomb barriers. ➤

### *New Scientist* May 25, 1996 Washington Watch

COLD fusion is the scientific embarrassment that won't go away. A small band of researchers across the US still tend their cold fusion cells, carefully noting how much electrical power goes in and how much heat comes out. Some are sponsored by Japanese research establishments, others are financed by private investors. Disregarding the scorn of the international physics community, they insist that their experiments still produce more heat than any chemical reaction could explain. Recently, several scientists at the renowned Naval Research Laboratory in Washington DC became intrigued by the experiments. They invited a few of the cold fusion researchers to present their work at the NRL. But, hearing of the meeting, higher-ranking officials at the laboratory sensed a PR fiasco in the making. They decided that cold fusion was still too hot to handle and banned media interviews with NRL scientists on the subject. The cold fusion meeting went on as scheduled, but inquiring reporters were informed that scientists at the NRL "had nothing to say" on the topic.

[thanks to Chris Tinsley and Arthur C. Clarke]

## COLD FUSION WRITER AT MIT

Cambridge, MA - Congratulations Gary Taubes, author of "Bad Science: The Short Life and Weird Times of Cold Fusion" who is one of eight journalists awarded the auspicious 1996-1997 Knight Fellowships at the Massachusetts Institute of Technology. This honor was originally named the Vannevar Bush Fellowships (inaugurated in 1983), but changed in 1987 when the Knight Foundation (Florida) became the principal sponsor. Judges included Jerry Bishop and five other editors, correspondents and reporters.



## Eight Journalists Awarded

### **Knight Fellowships** [MIT TechTalk - May 8, 1996]

Newspaper and magazine reporters of science, medicine and the environment from across the United States will join colleagues from Germany and Japan on September 1 as the 1996-97 Knight Science Journalism Fellows at MIT.

The journalists will spend nine months at the Institute in group and individual study. Among their group activities will be several field trips and approximately 55 seminars with faculty members on recent developments in science, technology, medicine and the environment. The new Fellows will be: William H. Allen, 44, science writer at the St. Louis Post-Dispatch.

Nancy R. Bazilchuk, 39, environment and science writer at The Burlington Free Press in Vermont.

Madeline Drexler, 41, of Watertown, medical columnist, The Boston Globe Magazine.

Ruth C. Flanagan, 31, of Madison WI, contributing editor for Earth magazine.

Mark A. Grossi, 42, environment reporter at The Fresno Bee in California, a McClatchy newspaper.

Carola Hanisch, 27, editor, economy, social affairs and science for Markische Allgemeine, Potsdam, Germany.

Asako Murakami, 32, environment staff writer on the national news desk of The Japan Times, Tokyo.

Gary A. Taubes, 40, of New York City, contributing correspondent to Science magazine and author of Nobel Dreams and Bad Science: The Short Life and Weird Times of Cold Fusion.

The Fellows will be the 14th group of visiting science journalists under a program inaugurated in 1983 as the Vannevar Bush Fellowships. The name was changed in 1987 when the John S. and James L. Knight Foundation of Miami became principal sponsor. The program is an activity of the Program in Science, Technology and Society in MIT's School of Humanities and Social Science.

The number of Fellows selected for the program now totals 145, representing 26 states, the District of Columbia and 15 foreign countries. Two Fellows have come from Australia, two from Brazil, three from Italy, three from Poland, four from Canada, nine from Germany and 13 from Japan.

Finalist interviews were conducted at MIT on April 26 and 27. The 1996 Selection Committee for the Knight Fellowships was: Jerry E. Bishop, deputy news editor at The Wall Street Journal; Deborah Fitzgerald, associate professor of the history of technology at MIT; Ann Gibbons, contributing correspondent to Science magazine; Robert Lee Hotz, science reporter at the Los Angeles Times, and Kathy Sawyer, science and space reporter at The Washington Post.

June 9, 1996

### Open Letter to President Vest:

Certain individuals within MIT have covered up the science field of cold fusion since 1989. These people continue to play havoc locally to alumni or professors who work in this field, and nationally interfere with its development. Harassment has included sabotaging normal teaching of the subject during IAP, ignoring requests made to the Presidential Office to correct this relevant matter, and possibly two attempts at stopping publications which are favorable to, or relate to, this field.

Is it MIT's choice to continue its name in association with a paper which deserves to be redacted? Or must MIT wait until history's Congressional Hearings note that nothing was done? Instead of correcting the matter, instead of even allowing discussion, to the consternation of scientists, engineers, alumni, and students, some staffers in President Vest's Office and others who took part in the cold fusion cover-up now have more clout than apparently does either TRUTH itself, or MIT alumni or Professors within MIT knocking upon the "locked door" of the President's Vest's Office.

We believe MIT should correct this matter and thereby rescue its reputation immediately. Continued denial by those in the Institute who know of this matter, coupled with a choking off of free thought and even classes involving this material and nuclear science and engineering is simply not acceptable.

Several concerned MIT Alumni





# COLD FUSION TIMES

## ENGINEERING AND MATERIALS VIEW TOWARD COLD FUSION

We have reported scores of reports confirming these systems for several years. Now in this issue there is confirmation from NASA, from EPRI, from Los Alamos, and others. This brief excerpt from the Clayton paper will corroborate what our readers have known for several years.

### TRITIUM PRODUCTION FROM A LOW VOLTAGE DEUTERIUM DISCHARGE ON PALLADIUM

T. N. Clayton, D. D. Jackson and D. G. Tuggle  
Los Alamos National Laboratory

Los Alamos, NM -- Over the past year we have been able to demonstrate that a plasma loading method produces an exciting and unexpected amount of tritium from small palladium wires. In contrast to electrochemical hydrogen or deuterium loading of palladium, this method yields a reproducible tritium generation rate when various electrical and physical conditions are met. Small diameter wires (100 - 250 microns) have been used with gas pressures above 200 torr at voltages and currents of about 2000 V at 3-5 A. By carefully controlling the sputtering rate of the wire, runs have been extended to hundreds of hours allowing a significant amount ( $> 10^5$  nCi) of tritium to accumulate. We will show tritium generation rates for deuterium-palladium foreground runs that are up to 25 times larger than hydrogen-palladium control experiments using materials from the same batch. We will illustrate the difference between batches of annealed palladium and as received palladium from several batches as well as the effect of other metals (Pt, Ni, Nb, Zr, V, W, Hf) to demonstrate

that the tritium generation rate can vary greatly from batch to batch. We will report on our tritium generation results from a palladium wire-plate configuration subjected to periodic pulsed deuterium or hydrogen plasma. This configuration is reproducible within a batch and produces a measurable amount of tritium in a few days. As in other work in this area, it has been found that the output is very batch dependent and sensitive to material impurities that prevent hydriding. As in our previous work<sup>1,3</sup>, all tritium data was obtained from several batches of 100 or 250 micron wire and 250 micron



Dr. N. Clayton

thick plate from J&M or Goodfellow metals. In these experiments most of the tritium data

was obtained with on-line tritium gas monitors. Several times, the gas was oxidized and tested with a scintillation counter. Some have criticized the detection of tritium because the signals seem to be (a) insignificant, (b) tritium is ubiquitous, and (c) the palladium metal is subject to possible tritium contamination. The magnitude of the signals discussed in this paper are multi-sigma and are sometimes over a hundred times the tritium background in the supply gas. Furthermore, the rate of tritium evolution in the sealed system may be the most sensitive and rapid indicator of anomalous nuclear behavior in deuterided metals.

The procedure for a plasma run was to first fill the 3.1 liter loop with deuterium gas at 600 torr and obtain a measure of the initial background tritium concentration. With the loop drift rate measured, the deuterium was circulated through the cell to slowly hydride the sample. The pressure in the cell and the loop was then lowered to the operating pressure by pumping the excess deuterium out. The wire was pulsed negatively, at 20 ms at 50 Hz, with currents between 2 and 5 A, voltages that varied from 1500 to 2500 V, and cell pressure of 300 torr. These conditions reduced the heating in the cell and maintained a cell to ambient temperature difference of less than 25°C to avoid gross dehydriding of the wire and plate. It appeared important to avoid a plasma condition that resulted in either a bright pink electron channel or arcing at the tip of the wire. After a few hours of plasma operation the voltage-current stabilized, presumably due to the formation of small cones (10-20 microns high) all over the surface of the wire. After 20 hours, palladium was visibly sputtered onto the plate. The sputter rate at 300 torr, 3.5 A, was about  $\sim 2$  Angstroms/s. The cell pressure was monitored, and if it did not drop after 24 hours (indicating hydriding), then a small amount of CO<sub>2</sub> (0.75% by vol) was added, which would initiate hydriding. ... If the reading was steady and CO<sub>2</sub> was added, then the gas was circulated through the liquid nitrogen cold trap to collect any water and determine if any tritiated water was present. The system was then pumped out, the cell closed off, and fresh deuterium added to the system after a couple of flushes with either fresh deuterium or air. The difference between the fresh deuterium and the deuterium reading after exposure to the plasma was used as the measure of the tritium content.

(continued next page)

### EPRI REPORT ON NUCLEAR PROCESSES IN DEUTERATED METALS

(continued from page 1) measured to an accuracy of approximately 0.5%. These excess powers integrated to a total of approximately 0.1 to 1.1 MJ for an approximate 2.5 g (1/40 mole) palladium cathode. Thus, the excess heats ranged between 4 to 44 MJ/mole of palladium, which was well above the largest known heats of chemical transformation in this or any other metal. The largest heat of chemical transformation in palladium is to the bromide at 0.9 MJ/mole. If the integrated excess powers are diluted by the electrochemically generated heat during the long initiation periods, net positive heat balances of 2 to 4% are obtained.

This work confirms the claims of Fleischmann, Pons, and Hawkins of the production of excess heat in deuterium-loaded palladium cathodes at levels too large for chemical transformation. However, the phenomena were obtained in only about half the cells. From the conditions of loading, initiation time, and current density on the successful observations of excess heat, it is understood why the phenomena are so difficult to attain. The conditions in the successful cells were not entirely under experimental control because the closed cells slowly leach silica and other materials from the anode and its supports as well as from the cell walls. This leached material can deposit on the cathode surface and interfere with the loading process. Also, the palladium purity depended on whatever was available from the manufacturer.

Subsequent research has shown a pronounced batch effect on successful loading from different shipments of palladium from the same as well as from different suppliers. It is suspected that metallurgical conditions as well as impurity content may be the source of this batch effect. The primary objective of further work on this subject will be to demonstrate which nuclear reactions, if any, are generating the excess heat. The only way to do this is to observe in at least roughly quantitative fashion the nuclear reaction products or "ashes." At this time, it is thought that the most likely ashes will be helium of mass 4 observable in the vapor phase of closed cells. The reaction producing helium needs to be known in order to maximize this excess heat phenomena for practical uses in the nuclear power industry.

[Ed.- The entire report on this Advanced Nuclear Technology is available through EPRI and is highly recommended.]

## コルト フュージョン タイムズ ANNOUNCEMENTS

### ICCF-6

October 13 (Sun.) - 18 (Fri.), 1996

- \* Excess Energy Phenomena in Deuterium/Metal Systems (Experiments and Theory)
- \* Correlation Between Excess Energy and Nuclear Products (Experiments and Theory)
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- \* Innovative Approaches

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### COLD FUSION TIMES

*The Journal Of The  
Scientific Aspects Of Loading  
Isotopic Fuels Into Materials*

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[The equipment used] one of two stainless steel gas analysis loops containing a 1.8 liter ion gauge and a 310.9 cc calibration volume. The atmospheric, ion gauge and sample pressure (0.2%), Femtotech and room temperatures (0.1°C) are recorded on a computer log at 60 s intervals. The pressure drop during hydriding of the wire and plate is used as an approximate indicator of the stoichiometry of the PdD<sub>x</sub>. Both loops have a heater to maintain the Femtotech (-0.03 nCi/°C) at a constant temperature, an integral cold trap, and there are valves that allow the pressurization of the cell independent of the loop. A two micron filter is installed at the inlet of the ion gauge and at the outlet of the cell to eliminate spurious responses due to particulates. To eliminate the possibility of oil contamination, a molecular drag and diaphragm pump is used to evacuate the system.

## TRITIUM PRODUCTION

(continued from page 6)

A total of 65 plasma wire experiments were performed, 12 of these were other than palladium wire and plate. Twenty experiments were run with multiple wires, usually 3 wires bundled together, and eight experiments used different thickness foils 25 to 125 microns thick. The balance of the tests were done with one 250 micron diameter wire and 250 micron thick plate. Three hydrogen plasma experiments were done with palladium plate and wire and two were done with platinum wire and plate. A summary of several background and foreground experiments is shown in Figure 2. [Ed. - on front page] The best experiment, produced 102 nCi.

The basic premise that the detected ionizing material is tritium is indisputable because, (a) quantitative measurements agree with the scintillation counter, (b) the gas may be transported on a clean palladium bed between different ionization systems and produces an increased reading commensurate with the decrease in tritium concentration noted in the initial system, (c) as the pressure is decreased the tritium signal is seen to decrease (for dry gas) in a manner consistent with the calibration for a known level of tritium in deuterium and finally, (d) the signal shows no diminution over a two week timeperiod, consistent with the half life of tritium.

Three types of contamination of the wire are possible; the first is just surface contamination due to atmospheric or liquid exposure to tritium, the second type might be a distributed impurity, and the third would be a spot contamination. To avoid surface contamination, we thoroughly clean and polish the palladium surface prior to each run. If it were still present, a surface contaminant would be immediately evident when the wire was introduced to the analysis loop and deuterided, but we have not seen evidence for this type of contamination. We attribute the residue and smoke seen from some of the wires to entrained lubricant due to drawing the wire. This lubricant tends to be drawn out and smeared throughout the length of the wire, which implies tritium contaminated oil should be detected in long sections of the wire. However, wires that showed obvious high levels of oil contamination did not show tritium, and we did not observe a large tritium signal when the wire was heated and the oil suddenly evolved. Similarly, the dark spots are present after simply heating the wire to 600°C but there is no evidence for tritium release at temperatures as high as 850°C.

An indicator that the tritium originated in the cells is that the output was sensitive to the metallurgical condition of the palladium. Palladium wire annealed in air showed a lower output than as received wire. Likewise palladium wire stress relieved but not annealed showed a similar (about a factor of 3) reduction in output. This could be interpreted as a release of tritium if it was contamination, however, then the tritium would have been easily detected in the heating controls. In sample #4, that showed significant tritium output, post heating (250°C) of the palladium wire released 12.4nCi of tritium.

continued next page)

## FOR THE WORLD

(More Information -- See Pages 2 And 3 Too)

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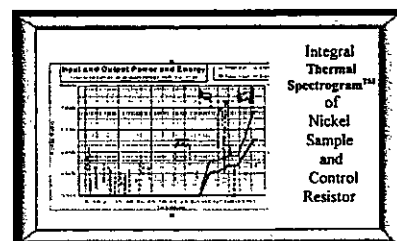
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# \*\* WHAT'S HAPPENIN' \*\*

# NEWS from the INTERNET

Summer '96

## CF MOVIE FIELD HEATS UP!

Prof Keith Johnson's Film Competes



MIT Professor Keith Johnson continues his computations on theoretical material science, movie productions ("Excess Heat"), and other interests in this pose for Joseph Mont, TAB Staff Writer (5/29/96)

HOLLYWOOD, CA; CAMBRIDGE, MA - As reported previously in the COLD FUSION TIMES, MIT Professor Keith Johnson has steadfastly continued production on his film: "Excess Heat". But suddenly, there is bigtime competition. At least two "cold fusion" genre films now exist.

Keanu Reeves appears in the first competing action film called "Chain Reaction" directed by Andrew Davis (*The Fugitive*). In the movie, Reeves plays also works as a machinist/inventor in the school's physics lab wherein he makes an invention involving "a new and cheap method of separating water into oxygen and hydrogen using electricity" to ultimately extract hydrogen energy from water - pollution free and cheap.



Keanu and his agent Rachel stand near their cold fusion apparatus (oblong tank is where the hydrogen energy/fusion occurs) during their nonstop chase across the US after being framed for murder! Keanu Charles Reeves (a.k.a. Chuck Spidena), born in Beirut ('64) to a geologist, is now in Amsterdam is playing a bass guitar with his pop band Dogstar. "The Wall" (his nickname) is back to his long haired image predating "A Walk In The Clouds", "Johnny Mnemonic", and "Speed". To "prepare" for his role, Keanu "spent some time with some physicists". No word on whether any were in cold fusion itself, so one can only presume Keanu heard the "hot fusion" side.

As in real life, the 'bad guys' simply do not want the world to know about this, but keep the invention for themselves. To this end, they pursue Keanu and his top physicist aide, English Lily Sinclair (played by Rachel Weisz), across the country after framing them for murder after the laboratory is destroyed by an explosion. 20th Century Fox plans a July 31st release. In the meantime, highly publicized previews with the Summer block-buster "Independence Day" are already drawing rave crowds. Props for CHAIN REACTION (outdated laboratory equipment) were reportedly bought for \$10,000 from the basement equipment stash at the University of Illinois.

Surprisingly, the film has received more attention and interest in *Science* than have the quality positive papers about this field by the same magazine.

"Researchers still have much to learn about sonoluminescence, a phenomenon in which small amounts of light are produced when sound waves are sent through water...In the film, Reeves watches, awestruck, as an electrified container of bubbling water puts out enough energy to run the world. Real-life scientists have found that the manipulation of sound frequencies can generate a stable bubble in water that may be as hot as 1 million Kelvin. But making the leap to usable energy production is a lot more problematic than Reeves makes it appear. If the film's premise is off the wall, the producers have at least taken pains to provide other dimensions of authenticity. ... Joseph Young, a physics graduate student at the University of Chicago who picked up a quick \$1000 for 3 days of work as science adviser for the film."

[Science, 19 April 1996]

## TRITIUM PRODUCTION (Continued From Page 8)

The fact that such high concentrations can be left in the palladium suggests that the process is near but not at the surface. The pulse length is sufficient for the diffusion of 200 Angstroms (10 ms pulse length) into the palladium. Then the tritium may be released when the surface layer of the palladium is sputtered by the energetic plasma. This would indicate that the tritium was in a 15 to 30 micron layer on the 250 micron in diameter wire. The fact that a significant amount of tritium shows up as (after the addition of CO<sub>2</sub>) TDO is also indicative of a near surface reaction. Dendrites and aspires (up to 20 microns high) on the surface of the palladium have been suggested as possible tritium formation sites....

We have found that the tritium output depends on the temperature, pressure and current applied to the cells. Yet, the tritium yield is most sensitive to the purity and metallurgical condition of palladium used in the experiments. Various tests for tritium contamination confirm that there is no initial tritium contamination in the powder, foil, wire or other materials used in this study. CO<sub>2</sub> additions had a remarkable effect on the production of tritium by these cells and the effect seems to be related to and enhancement of the hydriding of the palladium. It appears that very pure palladium is more effective than impure palladium in producing tritium. Based on our impurity analysis of the material we cannot identify a difference in concentration of a single impurity that is important to either include or exclude from the palladium. This is partially a morphological or metallurgical issue involving dislocations since we have seen a reduced output from annealed or stress relieved palladium when compared to as received palladium from the same batch. However, palladium that has been hydrided and dehydrided must always be annealed to reactivate it. The fact that most of the tritium is evolved promptly to the gas, yet significant amounts are found in the palladium suggest that the process is near but probably not at the surface.

1. Taggela, D. G., Cleyter, T. N., and Taylor, R. F.: Tritium Evolution from Cold Fusion, October 21-25 Nagoya Japan, Ed. H. Rogano, Universal Academy Press/Tokyo Japan, 1993, p217.
2. Taggela, D. G., Cleyter, T. N., and Taylor, R. F.: Tritium Evolution from Cold Fusion, October 21-25 Nagoya Japan, Ed. H. Rogano, Universal Academy Press/Tokyo Japan, 1993, p217.
3. Taggela, D. G., Cleyter, T. N., and Taylor, R. F.: Tritium Evolution from Cold Fusion, October 21-25 Nagoya Japan, Ed. H. Rogano, Universal Academy Press/Tokyo Japan, 1993, p217.
4. Cleyter, T. N., Taggela, D. G., Taylor, R. F., and Taylor, R. F.: Tritium Evolution from Cold Fusion, October 21-25 Nagoya Japan, Ed. H. Rogano, Universal Academy Press/Tokyo Japan, 1993, p217.
5. Taggela, D. G., Cleyter, T. N., and Taylor, R. F.: Tritium Evolution from Cold Fusion, October 21-25 Nagoya Japan, Ed. H. Rogano, Universal Academy Press/Tokyo Japan, 1993, p217.

## OP-ED COLUMN -- LETTERS

"Thanks for sending me the issue of *Cold Fusion Times*. Just for clarification, in vol 3, no. 1 on p.5 you quote me regarding a rule of thumb. A better statement would have been:

"A USEFUL RULE OF THUMB IS THAT IF THE SOLID IS COLORED OR BLACK, IT PROBABLY WON'T BE A GOOD ELECTROLYTE. A BLACK COLOR STRONGLY SUGGESTS THAT THERE IS A SIGNIFICANT ELECTRONIC CONDUCTIVITY. SINCE A GOOD ELECTROLYTE REQUIRES THAT THE ELECTRONIC COMPONENT OF THE CONDUCTIVITY BE MUCH LESS THAN THE IONIC COMPONENT, THIS MATERIAL WOULD NOT BE PRACTICAL AS AN ELECTROLYTE."

In principle, it could still be a good ionic conductor even if black; however, the electronic component would be expected to be high as well. In this case, this material would probably best be considered a mixed (ion and electron) conductor and not an electrolyte."

Hope this is useful. Aloha, Bruce Liebert Honolulu, Hawaii

Editor: Thanks. This Correction is duly noted.

What came first? Neither the chicken nor the egg are candidates. But the scripts of *Excess Heat* and *Chain Reaction* look similar in several regards. Ignoring the murders, cover-up, etc, the 20th Cent. Fox release indicates a system used for cheap, pollution free energy from water. To the right is the script from Johnson filed several years earlier, and which may have made its way to 20th Cent. Fox. Sound familiar, Art Buchwald?

CAROLAN  
But you said Professor Johnson was not a cold fusion.  
  
ALVIN  
She was my lab assistant when we discovered cold fusion.  
  
CAROLAN  
Unfortunately, she refused to accept the fact that cold fusion was just very bad science.  
  
CAROLAN  
Yes, but imagine, cheap, pollution-free energy from water. That's real. I'm all in. It's really weird.  
  
ALVIN  
Where's Professor Johnson now?

## WORLD WIDE WEB SOURCES ON COLD FUSION

• COLD FUSION TIMES' home page:  
<http://world.std.com/~mical/cf.html>

• John Logajon's home page:  
<http://www.skypoint.com/members/jlogajon>

• Infinite Energy home page:  
<http://www.mit.edu/8001/people/frei/CFdir/CFhome.html>  
<http://ourworld.compuserve.com/homepages/JedRothwell/>

TO COLDLY GO WHERE NO ONE HAS GONE BEFORE

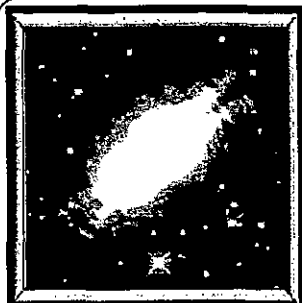
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## MORE REPORTS:

News and Developments from California, Washington, D.C., Illinois, Massachusetts, New Mexico, Rhode Island, Texas, Utah, China, England, France, India, Japan, Netherlands, Russia, Sri Lanka, United Kingdom. Plus copious material science and engineering news. Much has been going on.

We have reported scores of reports confirming cold fusion systems for several years. Now in this issue there is additional corroboration from NASA, EPRI, Los Alamos, and the March 1996 "normal temperature nuclear fusion" symposium held in China. This meeting was the latest in a series of Chinese symposia on cold fusion. This one reviewed the progress of a special fund established by the State (Chinese) Science and Technology Commission, and the Natural Science Foundation (China) to support cold fusion. Why the support? As Prof. Li has noted, "There are a lot of Chinese scientists who do not think there would be any chance for this phenomenon. They have not looked at any of the data since November 1989, and they veto any proposal on this research. Since the Natural Science Foundation in China operates on the basis of vote, it has been very difficult for "cold fusion" research to win the vote. However, a few judicious senior scientists with high reputation became aware of this situation, and made a suggestion

to the Natural Science Foundation in China and the State Science and Technology Commission. Based on this suggestion, a special fund was set up to support this research." The following important excerpt from New Energy News [Vol. 3, No. 10, April 1996] reviews the meeting.

## "NORMAL TEMPERATURE" FUSION SCIENCE NEWS FROM CHINA

Prof. Xing Zhong Li Dept. of Physics, Tsinghua Univ

Beijing, China -- The terminology "Normal Temperature Nuclear Fusion" reflected the Chinese view about the "cold fusion" phenomena. Most of us believe that it is a kind of nuclear fusion reaction, but it happens in the "normal temperature." In the Chinese language, "Normal Temperature" means that it is not high temperature (as high as that in the "hot fusion"); it is not low temperature (as low as that in the cryogenics for low temperature super conductor). There were about 30 conferees attending this symposium. Among them were the sponsors from the State Science and Technology Commission, and the Natural Science Foundation in China, a few senior scientists, the leaders of each research group, and some participants from each group. About one third of them were young graduate students and post-doctors. A major feature of this symposium was that the "excess heat" detection appeared in almost every research group. In the past five years, Chinese scientists paid more attention to nuclear signals (the charged particle detection, neutron, X-ray, and Gamma ray). (continued page 5)

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